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Chapter 1

Setup, Haar Measure, $L^2(G)$

1.1 Standing hypotheses

Definition 1 (Standing hypotheses for the compact group G). We fix once and for all a type G with the typeclass instances

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[Group G] [TopologicalSpace G] [IsTopologicalGroup G] [CompactSpace G] [T2Space G]
[MeasurableSpace G] [BorelSpace G].
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Throughout the blueprint, “ G ” refers to such a group.

Definition 2 (Standing hypotheses on the scalar field). We work over $\mathbb{C}$. (One may instead use `[RCLike k]` to cover real and complex cases simultaneously, but for clarity we fix $k = \mathbb{C}$.)

1.2 Haar measure and the probability normalization

Definition 3 (Haar probability measure μ_G). Let $\mu_G : \text{Measure } G$ denote the unique left-invariant regular Borel probability measure on G . Concretely: take Mathlib’s `MeasureTheory.Measure.haarMeasure` (with any compact set of positive measure as the normalization parameter) and rescale by $1/\mu(G)$ via `haarScalarFactor` so that $\mu_G(G) = 1$.

Theorem 4 (μ_G is a Haar measure). *$\mu_G.\text{IsHaarMeasure}$ holds.*

Theorem 5 (μ_G is a probability measure). *$\text{IsProbabilityMeasure } \mu_G$: $\mu_G(G) = 1$.*

Theorem 6 (μ_G is left-invariant). *$\mu_G.\text{IsMulLeftInvariant}$.*

Theorem 7 (μ_G is right-invariant). *$\mu_G.\text{IsMulRightInvariant}$. (Compact groups are unimodular; Mathlib’s `IsHaarMeasure` on a compact group derives this.)*

Theorem 8 (μ_G is inversion-invariant). *$\mu_G.\text{inv} = \mu_G$. (Mathlib lemma: `Measure.inv_eq_self_of_isMulRightInvariant` or equivalent.)*

1.3 The Hilbert space $L^2(G)$

Definition 9 ($L^2(G)$). $L^2(G) := L^2_{\mathbb{C}} \mu_G$, the Bochner L^2 space of $\mathbb{C}$ -valued square-integrable equivalence classes of functions on G , with inner product $\langle f, h \rangle := \int_G f(g) \overline{h(g)} d\mu_G(g)$.

Mathlib provides the structure (file `Mathlib.MeasureTheory.Function.L2Space`) including the inner-product instance `L2.innerProductSpace`, the norm $\|f\|_{L^2}^2 = \int |f|^2 d\mu_G$, and completeness (`MeasureTheory.Lp.completeSpace`).

Theorem 10 ($L^2(G)$ is a Hilbert space). *CompleteSpace $L^2(G)$.*

Theorem 11 ($C(G) \hookrightarrow L^2(G)$ as a continuous linear map). *There is a continuous linear map $\iota_{C \rightarrow L^2} : C(G, \mathbb{C}) \rightarrow L^2(G)$ taking a continuous function to its L^2 -equivalence class. Mathlib's `ContinuousMap.toLp` provides this.*

Theorem 12 ($\iota_{C \rightarrow L^2}$ has dense range). *The range of $\iota_{C \rightarrow L^2}$ is dense in $L^2(G)$. (Mathlib: `MeasureTheory.Lp.continuousMap_dense` or `ContinuousMap.toLp_denseRange` on a finite regular measure.)*

Theorem 13 ($\|\cdot\|_{L^2} \leq \|\cdot\|_{L^\infty}$ on $L^2(G)$). *For any $f \in C(G, \mathbb{C})$, $\|\iota_{C \rightarrow L^2}(f)\|_{L^2} \leq \|f\|_\infty$. (Follows from `MeasureTheory.MemLp.mono_exponent_of_isFiniteMeasure` or the direct integration estimate using $\mu_G(G) = 1$.)*

Chapter 2

Continuous Unitary Representations

This chapter develops the bundled type `UnitaryRep` and its basic API. Mathlib has `Representation k G V` (purely algebraic, no continuity) but does not have continuous unitary representations on general Hilbert spaces; everything in this chapter is [NEW].

2.1 The structure

Definition 14 (Continuous unitary representation). A *continuous unitary representation* of G on a complex Hilbert space H is a structure `UnitaryRep G H` bundling:

- `toMonoidHom : G →* (H →L[ℂ] H)`, the underlying group homomorphism into bounded operators;
- `isUnitary : ∀ g : G, IsUnitary (toMonoidHom g)`, where `IsUnitary T` is shorthand for `T.adjoint ∘L T = ContinuousLinearMap.id`;
- `strongContinuous : ∀ v : H, Continuous (fun g ⇒ toMonoidHom g v)`, strong-operator continuity.

We write $\rho : \text{UnitaryRep } G \ H$ for such an object and $\rho(g)v := \rho.\text{toMonoidHom } g \ v$.

Hypotheses on H throughout: `[NormedAddCommGroup H]` `[InnerProductSpace ℂ H]` `[CompleteSpace H]`.

Theorem 15 ($\rho(g)$ preserves inner products). For $\rho : \text{UnitaryRep } G \ H$, $g : G$, and $u, v : H$, $\langle \rho(g)u, \rho(g)v \rangle = \langle u, v \rangle$.

Proof. Compute $\langle \rho(g)u, \rho(g)v \rangle = \langle \rho(g)^* \rho(g)u, v \rangle$, then use `isUnitary` to replace $\rho(g)^* \rho(g)$ by `Id`. □

Theorem 16 ($\rho(g)$ is norm-preserving). $\|\rho(g)v\| = \|v\|$ for all g, v .

Proof. $\|\rho(g)v\|^2 = \langle \rho(g)v, \rho(g)v \rangle = \langle v, v \rangle = \|v\|^2$ by Theorem 15. □

Theorem 17 ($\rho(g)$ has operator norm ≤ 1). $\|\rho(g)\|_{\text{op}} \leq 1$.

Proof. By Theorem 16, $\|\rho(g)v\| \leq 1 \cdot \|v\|$ for every v , so $\|\rho(g)\|_{\text{op}} \leq 1$ by the definition of operator norm. □

2.2 Subrepresentations

Definition 18 (Predicate “closed subspace is ρ -invariant”). Given $\rho : \text{UnitaryRep } G \ H$ and a closed linear subspace V of H (Mathlib type `Submodule  $\mathbb{C}$   $H$`), define $\rho.\text{IsInvariant } V$ to mean: $\forall g : G, \forall v \in V, \rho(g)v \in V$.

Theorem 19 (Orthogonal complement of an invariant subspace is invariant). *Let V be a closed subspace of H with $\rho.\text{IsInvariant } V$. Then $\rho.\text{IsInvariant } V^\perp$.*

Proof. Let $w \in V^\perp$ and $g \in G$; we show $\rho(g)w \in V^\perp$. Pick $v \in V$. By Theorem 15, $\langle \rho(g)w, v \rangle = \langle w, \rho(g^{-1})v \rangle$. Since V is ρ -invariant by hypothesis, $\rho(g^{-1})v \in V$, so the right-hand side is 0 because $w \perp V$. Hence $\rho(g)w \perp v$ for every $v \in V$, i.e. $\rho(g)w \in V^\perp$. $\square$

Definition 20 (Subrepresentation as a structure). `UnitaryRep.Subrep  $\rho$` is the type of pairs (V, hV) where V is a closed subspace of H and $\text{hV} : \rho.\text{IsInvariant } V$.

Definition 21 (Restriction $\rho|_V$). Given $W : \text{UnitaryRep.Subrep } \rho$, the restriction $\rho|_W : \text{UnitaryRep } G \ W$ is given by $(\rho|_W)(g)(v) := \langle \rho(g)v, \cdot \rangle_W$, with the inherited Hilbert-space structure on W as a closed subspace. Strong continuity, unitarity, and homomorphism property are inherited.

2.3 Irreducibility

Definition 22 (Irreducible representation, predicate form). $\rho.\text{IsIrreducible}$ is the conjunction of

- `nontrivial` : `Nontrivial  $H$` , equivalently $H \neq 0$;
- `minimal` : $\forall V : \text{Submodule } \mathbb{C} \ H, \text{IsClosed } V \rightarrow \rho.\text{IsInvariant } V \rightarrow V = \perp \vee V = \top$.

2.4 Intertwiners and equivalences

Definition 23 (Intertwiner predicate). Let $\rho : \text{UnitaryRep } G \ H$ and $\sigma : \text{UnitaryRep } G \ K$. A bounded linear map $T : H \rightarrow_L K$ is an *intertwiner* (`$T.\text{IsIntertwiner } \rho \ \sigma$`) if $\forall g : G, T \circ_L \rho(g) = \sigma(g) \circ_L T$.

Definition 24 (Space of intertwiners). `UnitaryRep.Intertwiner  $\rho \ \sigma$` is the bundled type $\{T : H \rightarrow_L K \mid T.\text{IsIntertwiner } \rho \ \sigma\}$, registered as a $\mathbb{C}$ -module subspace of $H \rightarrow_L K$.

Definition 25 (Predicate “ T is a unitary equivalence”). An intertwiner $T : H \rightarrow_L K$ is a *unitary equivalence* if it is also a surjective isometry, equivalently $T^*T = \text{Id}_H$ and $TT^* = \text{Id}_K$.

Definition 26 (Equivalence of representations). Two representations ρ, σ are *equivalent*, $\rho \cong \sigma$, if there exists a $T : H \rightarrow_L K$ with $T.\text{IsIntertwiner } \rho \ \sigma$ and $T.\text{IsEquiv}$.

2.5 The left and right regular representations

Definition 27 (Left translation as a map on L^2). For $g \in G$, the left translation $\lambda_g : L^2(G) \rightarrow L^2(G)$ is the unique continuous linear map sending the class of $f \in C(G)$ to the class of $h \mapsto f(g^{-1}h)$. Existence: on continuous functions left translation preserves the L^2 -norm by left-invariance of μ_G (Theorem 6), hence extends uniquely by continuity to all of $L^2(G)$ (Theorem 12).

Theorem 28 (λ_g is unitary). *For each $g \in G$, λ_g is a unitary operator on $L^2(G)$.*

Proof. By construction λ_g preserves $\|\cdot\|_{L^2}$ on the dense subspace of continuous functions. Continuity extends this to all of $L^2(G)$. The inverse is $\lambda_{g^{-1}}$. $\square$

Theorem 29 (λ is a group homomorphism on L^2). $\lambda_{gh} = \lambda_g \circ_L \lambda_h$ and $\lambda_1 = \text{Id}$.

Proof. Verify on the dense subspace of continuous functions, where $f((gh)^{-1}x) = f(h^{-1}g^{-1}x) = (\lambda_h f)(g^{-1}x)$. Continuity extends to all of L^2 . $\square$

Theorem 30 (The orbit map of left translation is continuous). *For every $f \in L^2(G)$, the map $g \mapsto \lambda_g f$ is continuous from G to $L^2(G)$.*

Proof. Three steps.

Step 1: continuous functions are uniformly continuous on G . Mathlib's `CompactSpace.uniformContinuous_of_continuous` applied to $f \in C(G, \mathbb{C})$.

Step 2: convergence on continuous f . For $f \in C(G)$ and $\varepsilon > 0$, uniform continuity gives a neighbourhood U of 1_G with $|f(u^{-1}h) - f(h)| < \varepsilon$ for all $u \in U, h \in G$. Then $\|\lambda_u f - f\|_{L^2}^2 \leq \varepsilon^2$ since $\mu_G(G) = 1$. By group homomorphism (Theorem 29), $g \mapsto \lambda_g f$ is continuous at every $g \in G$.

Step 3: extend to all $f \in L^2$. Let $f \in L^2(G)$, $\varepsilon > 0$. Pick $f' \in C(G)$ with $\|f - f'\|_{L^2} < \varepsilon/3$ (Theorem 12). By Step 2, pick a neighbourhood U of g in G with $\|\lambda_h f' - \lambda_g f'\|_{L^2} < \varepsilon/3$ for $h \in U$. Triangle inequality with $\|\lambda_h\|_{\text{op}} = 1$ (Theorem 28) gives the bound for f . $\square$

Definition 31 (The left regular representation ρ^L). The left regular representation $\rho^L : \text{UnitaryRep } G \rightarrow L^2(G)$ is the bundling of $g \mapsto \lambda_g$ together with Theorems 28, 29, 30.

Definition 32 (Right translation as a map on L^2). For $g \in G$, $\rho_g : L^2(G) \rightarrow L^2(G)$ is the unique continuous linear extension of $f \mapsto (h \mapsto f(hg))$ from $C(G)$.

Definition 33 (The right regular representation ρ^R). Bundled in the same way as ρ^L , using Theorem 7 (right-invariance) in place of Theorem 6.

Theorem 34 (ρ^L and ρ^R commute). *For all $g, h \in G$, $\rho^L(g) \circ_L \rho^R(h) = \rho^R(h) \circ_L \rho^L(g)$.*

Proof. On the dense subspace of continuous functions: $(\rho^L(g)\rho^R(h)f)(x) = f(g^{-1}xh) = (\rho^R(h)\rho^L(g)f)(x)$, using associativity of the group operation. Continuity extends to all of L^2 . $\square$

Chapter 3

Convolution as a Bounded Operator on $L^2(G)$

The goal of this chapter is to define, for $\phi \in L^2(G)$, a bounded linear operator $T_\phi : L^2(G) \rightarrow L^2(G)$ given pointwise (on a dense subspace) by $T_\phi f = f * \phi$, and to identify three properties of T_ϕ : compactness, self-adjointness when ϕ is symmetric, and right-equivariance.

3.1 Pointwise convolution

Definition 35 (Pointwise convolution). For Borel-measurable $f, \phi : G \rightarrow \mathbb{C}$ such that the integral below converges, define

$$(f * \phi)(x) := \int_G f(y) \phi(y^{-1}x) d\mu_G(y).$$

In Lean this is `MeasureTheory.mconvolution` applied to the bilinear form $(\xi, \eta) \mapsto \xi \cdot \eta : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$.

Theorem 36 (Cauchy–Schwarz bound for pointwise convolution). For $f, \phi \in L^2(G)$ and $x \in G$, $|(f * \phi)(x)| \leq \|f\|_{L^2} \|\phi\|_{L^2}$.

Proof. $|(f * \phi)(x)| \leq \int |f(y)| |\phi(y^{-1}x)| d\mu_G(y) \leq \|f\|_{L^2} \cdot \|y \mapsto \phi(y^{-1}x)\|_{L^2}$ by the standard Cauchy–Schwarz inequality. Right-invariance and inversion-invariance of μ_G (Theorems 7, 8) give $\|y \mapsto \phi(y^{-1}x)\|_{L^2} = \|\phi\|_{L^2}$. $\square$

Theorem 37 ($f * \phi$ is continuous when $f, \phi \in L^2$). For $f, \phi \in L^2(G)$, the function $x \mapsto (f * \phi)(x)$ is continuous on G .

Proof. Write $(f * \phi)(x) = \langle f, L^2\text{-class of } (y \mapsto \phi(y^{-1}x)^*) \rangle_{L^2}$. The map $x \mapsto (y \mapsto \phi(y^{-1}x)^*)$ is continuous from G to $L^2(G)$ (Theorem 30 applied to the conjugate of the inverted reflection of ϕ). Continuity of the inner product gives the result. $\square$

3.2 Convolution as a bounded operator

Theorem 38 (L^2 norm bound for convolution). For $f, \phi \in L^2(G)$, $\|f * \phi\|_{L^2} \leq \|f\|_{L^2} \|\phi\|_{L^2}$ (using $\mu_G(G) = 1$).

Proof. Square the pointwise bound from Theorem 36 and integrate; use $\mu_G(G) = 1$. $\square$

Definition 39 (Bounded operator T_ϕ). For $\phi \in L^2(G)$, define $T_\phi : L^2(G) \rightarrow_L L^2(G)$ as the unique continuous linear extension of the pointwise-convolution operation $f \mapsto f * \phi$ from the subspace where pointwise convolution is well-defined, with norm bound $\|T_\phi\|_{\text{op}} \leq \|\phi\|_{L^2}$ from Theorem 38.

3.3 Right-equivariance of T_ϕ

Theorem 40 (T_ϕ commutes with right translation). For all $g \in G$ and $\phi \in L^2(G)$, $T_\phi \circ_L \rho^R(g) = \rho^R(g) \circ_L T_\phi$.

Proof. Sufficient to verify on continuous f , where

$$\begin{aligned} (T_\phi(\rho^R(g)f))(x) &= \int f(yg) \phi(y^{-1}x) d\mu_G(y) \\ &= \int f(z) \phi(z^{-1}gx) d\mu_G(z) \quad (z = yg) \\ &= (T_\phi f)(gx) = (\rho^R(g)T_\phi f)(x), \end{aligned}$$

using right-invariance (Theorem 7). $\square$

Corollary 41 (T_ϕ is an intertwiner $\rho^R \rightarrow \rho^R$). $T_\phi.$ *IsIntertwiner* $\rho^R \rho^R$.

Proof. Direct from the definition of intertwiner; the equation in Theorem 40 is the intertwiner condition. $\square$

3.4 Self-adjointness of T_ϕ for symmetric ϕ

Definition 42 (Symmetric kernel). $\phi \in L^2(G)$ is *symmetric* if $\phi(g) = \overline{\phi(g^{-1})}$ for μ_G -a.e. g .

Theorem 43 (Adjoint of T_ϕ). For every $\phi \in L^2(G)$ and $f, h \in L^2(G)$, $\langle T_\phi f, h \rangle = \langle f, T_{\tilde{\phi}} h \rangle$, where $\tilde{\phi}(g) := \overline{\phi(g^{-1})}$. Equivalently, $(T_\phi)^* = T_{\tilde{\phi}}$.

Proof. Compute via Fubini: $\langle T_\phi f, h \rangle = \int \int f(y) \phi(y^{-1}x) \overline{h(x)} d\mu_G(y) d\mu_G(x)$, swap the order, change variable $x \mapsto yx$ using right-invariance, then use $\phi(z) = \tilde{\phi}(z^{-1})$ and inversion-invariance (Theorem 8) to arrive at $\langle f, T_{\tilde{\phi}} h \rangle$. $\square$

Corollary 44 (T_ϕ self-adjoint when ϕ symmetric). If ϕ is symmetric (Definition 42), then T_ϕ is self-adjoint, i.e. $T_\phi^* = T_\phi$.

Proof. Symmetry says $\tilde{\phi} = \phi$ a.e., so by Theorem 43 $T_\phi^* = T_{\tilde{\phi}} = T_\phi$. $\square$

3.5 Compactness of T_ϕ

Theorem 45 (T_ϕ is a compact operator). For every $\phi \in L^2(G)$, the operator T_ϕ is compact in the sense of Mathlib's *IsCompactOperator*.

Proof. Routine but technical: T_ϕ is a Hilbert–Schmidt integral operator with kernel $K(x, y) = \phi(y^{-1}x)$. Its Hilbert–Schmidt norm is $\|K\|_{L^2(G \times G)}^2 = \int \int |\phi(y^{-1}x)|^2 = \|\phi\|_{L^2}^2$ (by right- and inversion-invariance, Theorems 7, 8). Hilbert–Schmidt operators are compact in any Hilbert space.

In Lean, this requires either (i) the abstract result “Hilbert–Schmidt operators are compact” applied to K , or (ii) a direct proof via approximation by finite-rank operators using density of $C(G \times G)$ in $L^2(G \times G)$ and continuity of $\phi \mapsto T_\phi$ (norm bound from Theorem 38). Approach (ii) avoids needing Hilbert–Schmidt theory and is preferred.

Sketch of (ii) for the formaliser:

1. For $\phi \in C(G)$, the operator T_ϕ has finite rank after approximation: choose a partition of unity to write $\phi = \sum_i c_i \mathbf{1}_{A_i}$ approximately, giving $T_\phi \approx \sum_i c_i$ (rank-1 piece).
2. Continuous functions are dense in $L^2(G)$, so T_ϕ for $\phi \in L^2$ is the operator-norm limit of finite-rank operators (using the bound $\|T_{\phi_1} - T_{\phi_2}\|_{\text{op}} \leq \|\phi_1 - \phi_2\|_{L^2}$ from Definition 39).
3. Compact operators form a closed subspace of bounded operators (Mathlib’s `isClosed_setOf_isCompactOperator`). Hence T_ϕ is compact.

□

Chapter 4

Approximate Identities

We need a family of symmetric continuous bumps $\{\phi_U\}$ supported in shrinking neighbourhoods of 1_G , with $T_{\phi_U} \rightarrow \text{Id}$ strongly as $U \rightarrow \{1_G\}$.

Definition 46 (Continuous bump). Given an open neighbourhood U of 1_G , a function $\phi : G \rightarrow \mathbb{R}$ is a *bump on U* if it is continuous, non-negative, has support contained in U , satisfies $\int_G \phi d\mu_G = 1$, and $\phi(g) = \phi(g^{-1})$ for all g .

Theorem 47 (Existence of continuous bumps). *For every open $U \ni 1_G$, there exists $\phi : G \rightarrow \mathbb{R}$ that is a bump on U .*

Proof. Use Urysohn's lemma in the locally compact Hausdorff group G to obtain a continuous non-negative $\psi \in C(G)$ with $\psi(1_G) > 0$ and $\text{supp } \psi \subseteq U$. Pick a smaller open $V \ni 1_G$ with $V \cup V^{-1} \subseteq U$ and shrink ψ 's support if necessary. Symmetrize: set $\psi^*(g) := \psi(g) + \psi(g^{-1})$. Then ψ^* has support in $V \cup V^{-1} \subseteq U$ and is symmetric. Normalise: $\phi := \psi^* / \int \psi^* d\mu_G$ (positive denominator because $\psi^*(1_G) > 0$ and ψ^* continuous non-negative implies a positive integral on any non-empty open set by the open-positivity of μ_G). $\square$

Theorem 48 (Convergence of bump-convolution to identity, L^2). *Let $\{U_n\}$ be a countable neighbourhood basis at 1_G shrinking to $\{1_G\}$, and ϕ_n a bump on U_n for each n . Then for every $f \in L^2(G)$, $T_{\phi_n} f \rightarrow f$ in L^2 as $n \rightarrow \infty$.*

Proof. Three steps.

Step 1. For continuous $f \in C(G)$, by uniform continuity (compactness, Heine–Cantor) of f , given $\varepsilon > 0$ pick a neighbourhood $U \ni 1_G$ with $|f(u^{-1}x) - f(x)| < \varepsilon$ for all $u \in U, x \in G$. For n large enough ϕ_n has support in U , so $|(T_{\phi_n} f)(x) - f(x)| = |\int (f(y^{-1}x) - f(x))\phi_n(y) d\mu_G(y)| \leq \varepsilon$. Hence $\|T_{\phi_n} f - f\|_{L^\infty} \rightarrow 0$, and a fortiori $\|T_{\phi_n} f - f\|_{L^2} \rightarrow 0$ since $\|\cdot\|_{L^2} \leq \|\cdot\|_{L^\infty}$ (Theorem 13).

Step 2. For general $f \in L^2(G)$ and $\varepsilon > 0$, pick $f' \in C(G)$ with $\|f - f'\|_{L^2} < \varepsilon/3$ (Theorem 12). By Step 1, pick N with $\|T_{\phi_n} f' - f'\|_{L^2} < \varepsilon/3$ for $n \geq N$.

Step 3. Triangle inequality:

$$\|T_{\phi_n} f - f\|_{L^2} \leq \|T_{\phi_n} (f - f')\|_{L^2} + \|T_{\phi_n} f' - f'\|_{L^2} + \|f' - f\|_{L^2}.$$

The middle term is $< \varepsilon/3$ by choice of N . The first and third are each $< \varepsilon/3$ using $\|T_{\phi_n}\|_{\text{op}} \leq \|\phi_n\|_{L^2}$ (Definition 39)... but $\|\phi_n\|_{L^2}$ is not bounded in general! We need a refined estimate: $\|T_{\phi_n} (f - f')\|_{L^2} \leq \|f - f'\|_{L^2} \cdot \|\phi_n\|_{L^1} = \|f - f'\|_{L^2}$ using $\|\phi_n\|_{L^1} = 1$ (the normalization condition) and Young's inequality with $L^2 \cdot L^1 \rightarrow L^2$. This bound is uniform in n . Then the first term is $< \varepsilon/3$. Total: $< \varepsilon$. $\square$

Chapter 5

The Compact Self-Adjoint Spectral Theorem (External)

This chapter contains a single external assumption: the spectral theorem for compact self-adjoint operators on a complex Hilbert space. This is not currently in Mathlib (Jan 2026 cutoff); the formalisation must either supply it or assume it as a hypothesis.

Assumption 49 (Spectral theorem for compact self-adjoint operators). *Let H be a complex Hilbert space and $T : H \rightarrow_L H$ a compact self-adjoint operator (in the sense of `IsCompactOperator` T and `IsSelfAdjoint` T). Then there exists an at-most-countable index set I , a function $\lambda : I \rightarrow \mathbb{R} \setminus \{0\}$ (the non-zero eigenvalues), and for each $i \in I$ a non-zero closed subspace $H_i := \ker(T - \lambda_i \text{Id})$ such that:*

1. *each H_i is finite-dimensional;*
2. *the family $\{H_i\}_{i \in I}$ is mutually orthogonal;*
3. *if I is infinite, the sequence $|\lambda_i|$ tends to 0;*
4. *the closure of the linear span $\bigsqcup_i H_i$ equals $(\ker T)^\perp$;*
5. *equivalently, $H = \ker T \oplus \widehat{\bigoplus_i H_i}$ as a Hilbert sum.*

Theorem 50 (Existence of a non-zero eigenvalue when $T \neq 0$). *Let $T : H \rightarrow_L H$ be compact, self-adjoint, and non-zero. Then there exists $\lambda \in \mathbb{R} \setminus \{0\}$ and a non-zero finite-dimensional closed subspace $H_\lambda \leq H$ with $T|_{H_\lambda} = \lambda \cdot \text{Id}$.*

Proof. By Assumption 49, $H = \ker T \oplus \widehat{\bigoplus_i H_i}$. If $T \neq 0$, then T is non-zero on $(\ker T)^\perp$, so the family $\{H_i\}_i$ is non-empty (otherwise the closure of their span would be $\{0\} \neq (\ker T)^\perp$). Pick any i ; take $\lambda := \lambda_i$ and $H_\lambda := H_i$. □

Chapter 6

Schur's Lemma for Continuous Unitary Irreducibles

6.1 Decomposition into self-adjoint pieces

Definition 51 (Self-adjoint and skew parts of an operator). For $T : H \rightarrow_L H$ on a complex Hilbert space, define $\operatorname{Re}(T) := (T + T^*)/2$ and $\operatorname{Im}(T) := (T - T^*)/(2i)$, both self-adjoint, with $T = \operatorname{Re}(T) + i \operatorname{Im}(T)$.

Theorem 52 (Equivariance descends to real and imaginary parts). If $T : H \rightarrow_L H$ has $T.\operatorname{IsIntertwiner} \rho \rho$, then so do $\operatorname{Re}(T)$ and $\operatorname{Im}(T)$.

Proof. Equivariance for T^* : from $T\rho(g) = \rho(g)T$, take adjoints to get $\rho(g)^*T^* = T^*\rho(g)^*$. By unitarity (Theorem 15), $\rho(g)^* = \rho(g^{-1})$. So $T^*\rho(g^{-1}) = \rho(g^{-1})T^*$ for every g , equivalently T^* intertwines ρ with itself. Linear combinations preserve the intertwining property. $\square$

6.2 Self-adjoint equivariant operator on irreducible is scalar

Theorem 53 (Eigenspaces of an equivariant operator are invariant). Let $T : H \rightarrow_L H$ be such that $T.\operatorname{IsIntertwiner} \rho \rho$, and $\lambda \in \mathbb{C}$. Then $\rho.\operatorname{IsInvariant} \ker(T - \lambda \operatorname{Id})$.

Proof. If $Tv = \lambda v$ then $T(\rho(g)v) = \rho(g)(Tv) = \lambda \rho(g)v$. $\square$

Theorem 54 (Self-adjoint equivariant compact operator on an irreducible has a single non-zero eigenvalue or is zero). Let $\rho : \operatorname{UnitaryRep} G H$ be irreducible, and $A : H \rightarrow_L H$ compact, self-adjoint, with $A.\operatorname{IsIntertwiner} \rho \rho$. Then either $A = 0$, or there exists $\lambda \in \mathbb{R} \setminus \{0\}$ with $A = \lambda \cdot \operatorname{Id}$.

Proof. If $A \neq 0$, by Theorem 50 there exist $\lambda \neq 0$ real and a non-zero finite-dimensional $H_\lambda := \ker(A - \lambda \operatorname{Id})$. By Theorem 53 H_λ is closed and ρ -invariant. By irreducibility of ρ , the only closed invariant subspaces are $\perp$ and $\top$; since H_λ is non-zero, $H_\lambda = \top$, i.e. $H_\lambda = H$. Thus $Av = \lambda v$ for every $v \in H$, i.e. $A = \lambda \cdot \operatorname{Id}$. $\square$

6.3 Schur's lemma proper

Theorem 55 (Schur's lemma, scalar form, compact case). *Suppose $\rho : \text{UnitaryRep } G \ H$ is irreducible, and let $T : H \rightarrow_L H$ be compact with $T.\text{IsIntertwiner } \rho \ \rho$. Then there exists $z \in \mathbb{C}$ with $T = z \cdot \text{Id}$.*

Proof. Apply Theorem 52 to obtain $\text{Re}(T), \text{Im}(T)$ both self-adjoint and equivariant. Their compactness follows from compactness of T and T^* (compact operators are closed under adjoint and linear combinations: Mathlib's `IsCompactOperator.add`, `IsCompactOperator.smul`, `IsCompactOperator.adjoint`). By Theorem 54, each of $\text{Re}(T), \text{Im}(T)$ is a real scalar (or zero); their combination $T = \text{Re}(T) + i \text{Im}(T)$ is a complex scalar. $\square$

Remark 1. *This Schur is weaker than the classical statement (which removes the compactness hypothesis on T): we use compactness of T to invoke the spectral theorem (Assumption 49). The full Schur for arbitrary bounded equivariant T requires the spectral theorem for bounded self-adjoint operators (a bigger black box), but in our application every intertwiner we will encounter is compact (it factors through finite-dimensional or compact-operator endpoints).*

Theorem 56 (Schur for non-equivalent compact intertwiner targets). *Let $\rho : \text{UnitaryRep } G \ H$, $\sigma : \text{UnitaryRep } G \ K$ both irreducible, and $T : H \rightarrow_L K$ compact with $T.\text{IsIntertwiner } \rho \ \sigma$. If $\rho \not\cong \sigma$ (no unitary equivalence between ρ and σ), then $T = 0$.*

Proof. Suppose $T \neq 0$. Since $\overline{\ker T}$ is closed and ρ -invariant (intertwiner property gives invariance, then take closure), and ρ is irreducible, $\overline{\ker T} \in \{0, H\}$. If $\overline{\ker T} = H$, density gives $T = 0$, contradiction. So $\overline{\ker T} = 0$ (the kernel itself is closed for bounded T already), i.e. T is injective.

Similarly $\overline{T(H)}$ is closed and σ -invariant, so $\overline{T(H)} \in \{0, K\}$ by irreducibility of σ . The case $\overline{T(H)} = 0$ gives $T = 0$, contradiction; so $\overline{T(H)} = K$, i.e. T has dense range.

Consider $T^*T : H \rightarrow_L H$. It is self-adjoint, positive, ρ -equivariant, and compact (compact operators absorb bounded ones: T compact $\Rightarrow T^*T$ compact). By Theorem 55, $T^*T = c \cdot \text{Id}$ for some $c \in \mathbb{C}$. Positivity and $T \neq 0$ force $c > 0$.

Set $U := T/\sqrt{c} : H \rightarrow_L K$. Then $U^*U = \text{Id}_H$, so U is an isometry. Density of $T(H)$ in K implies density of $U(H)$ in K , hence (since the image of an isometry is closed) $U(H) = K$. So U is a unitary equivalence $\rho \rightarrow \sigma$, contradicting $\rho \not\cong \sigma$. $\square$

Chapter 7

Matrix Coefficients

7.1 Matrix coefficient of a finite-dim representation

Definition 57 (Matrix coefficient $\pi_{u,v}$). Let V be a finite-dimensional complex Hilbert space and $\rho : \text{UnitaryRep } G \rightarrow V$. For $u, v \in V$, define $\pi_{u,v}^\rho : G \rightarrow \mathbb{C}$ by $\pi_{u,v}^\rho(g) := \langle \rho(g)u, v \rangle_V$.

Theorem 58 (Matrix coefficients are continuous). $\pi_{u,v}^\rho : G \rightarrow \mathbb{C}$ is continuous.

Proof. Composition of (i) $g \mapsto \rho(g)u$ continuous from G to V (strong continuity, Definition 14 third field); (ii) $w \mapsto \langle w, v \rangle$ continuous (Cauchy–Schwarz). $\square$

Theorem 59 (Matrix coefficients are bounded). $|\pi_{u,v}^\rho(g)| \leq \|u\| \|v\|$.

Proof. Cauchy–Schwarz plus $\|\rho(g)u\| = \|u\|$ from Theorem 16. $\square$

Corollary 60 (Matrix coefficients are in $C(G)$ and $L^2(G)$). $\pi_{u,v}^\rho$ defines an element of $C(G, \mathbb{C})$ (via Theorem 58 bundling continuity) and via $\iota_{C \rightarrow L^2}$ an element of $L^2(G)$.

7.2 Algebra structure

Theorem 61 (Sum of matrix coefficients is a matrix coefficient). Let V_1, V_2 be finite-dim Hilbert spaces with reps ρ_1, ρ_2 . For $u_i, v_i \in V_i$, the function $g \mapsto \pi_{u_1, v_1}^{\rho_1}(g) + \pi_{u_2, v_2}^{\rho_2}(g)$ equals $\pi_{u_1 \oplus u_2, v_1 \oplus v_2}^{\rho_1 \oplus \rho_2}$ on $V_1 \oplus V_2$.

Proof. Direct expansion of inner product on a direct sum. $\square$

Theorem 62 (Scalar multiple of a matrix coefficient is a matrix coefficient). $c \cdot \pi_{u,v}^\rho = \pi_{cu,v}^\rho$ for $c \in \mathbb{C}$.

Proof. $\langle \rho(g)(cu), v \rangle = c \langle \rho(g)u, v \rangle$. $\square$

Definition 63 (Tensor product of unitary reps). Given $\rho_i : \text{UnitaryRep } G \rightarrow V_i$ for $i = 1, 2$ with V_i finite-dimensional, $\rho_1 \otimes \rho_2 : \text{UnitaryRep } G \rightarrow (V_1 \otimes V_2)$ is the homomorphism $g \mapsto \rho_1(g) \otimes \rho_2(g)$ on the algebraic (= Hilbert, in finite dim) tensor product, with the inner product $\langle u_1 \otimes u_2, v_1 \otimes v_2 \rangle := \langle u_1, v_1 \rangle \langle u_2, v_2 \rangle$ extended linearly.

Theorem 64 (Product of matrix coefficients). $\pi_{u_1, v_1}^{\rho_1}(g) \cdot \pi_{u_2, v_2}^{\rho_2}(g) = \pi_{u_1 \otimes u_2, v_1 \otimes v_2}^{\rho_1 \otimes \rho_2}(g)$.

Proof. $\langle \rho_1(g)u_1, v_1 \rangle \langle \rho_2(g)u_2, v_2 \rangle = \langle (\rho_1 \otimes \rho_2)(g)(u_1 \otimes u_2), v_1 \otimes v_2 \rangle$ by definition of tensor product inner product. $\square$

Definition 65 (Contragredient (dual) representation). Given $\rho : \mathbf{UnitaryRep} \ G \ V$ with V finite-dimensional, $\bar{\rho} : \mathbf{UnitaryRep} \ G \ V^*$ is the homomorphism $g \mapsto (\rho(g)^{-1})^* = \rho(g^{-1})^* = \rho(g)$ acting on the conjugate-dual space V^* via $(\bar{\rho}(g)\ell)(v) := \overline{\ell(\rho(g)^{-1}v)}$; equivalently we identify V^* with the conjugate Hilbert space $\overline{V}$ (same underlying set, scalar multiplication conjugated).

Theorem 66 (Conjugate of a matrix coefficient). $\overline{\pi_{u,v}^\rho(g)} = \pi_{v,u}^{\bar{\rho}}(g)$ (under the conjugate-Hilbert identification).

Proof. $\overline{\langle \rho(g)u, v \rangle} = \langle v, \rho(g)u \rangle = \langle \rho(g^{-1})v, u \rangle = \pi_{v,u}^{\bar{\rho}}(g)$ using unitarity. $\square$

Theorem 67 (The constant function 1 is a matrix coefficient). Let $\mathbf{1} : \mathbf{UnitaryRep} \ G \ \mathbb{C}$ be the trivial representation $\mathbf{1}(g) = \text{Id}_{\mathbb{C}}$. Then $\pi_{1,1}^{\mathbf{1}}(g) = 1$ for all g .

Proof. $\langle 1, 1 \rangle_{\mathbb{C}} = 1$ in the trivial inner product. $\square$

7.3 The matrix-coefficient star-subalgebra

Definition 68 (The matrix-coefficient subset of $C(G)$). $\mathcal{M} := \{\pi_{u,v}^\rho \mid (\rho, V) \text{ a finite-dim continuous unitary rep of } G, \text{ and } u, v \in V\} \subseteq C(G, \mathbb{C})$.

Definition 69 (The matrix-coefficient star-subalgebra). $\mathcal{A} \subseteq C(G, \mathbb{C})$ is the **StarSubalgebra** $\mathbb{C} \langle \mathcal{M}, \mathbb{C} \rangle$ generated by $\mathcal{M}$. By Theorems 61, 62, 64, 66, 67, $\mathcal{A} = \text{span}_{\mathbb{C}} \mathcal{M}$ (already closed under multiplication and complex conjugation), and contains the constants.

Chapter 8

Existence of Separating Finite-Dim Irreducibles

This is the technical heart of Tao's proof. We use the spectral theorem on convolution operators to build a finite-dimensional irreducible subrepresentation of $L^2(G)$ on which a chosen $g_0 \neq 1_G$ acts non-trivially.

8.1 Setup and key intermediate lemmas

Theorem 70 (Right regular action is faithful). *For every $g_0 \in G$ with $g_0 \neq 1_G$, there exists $f \in L^2(G)$ with $\rho^R(g_0)f \neq f$.*

Proof. Pick disjoint open neighbourhoods $U \ni 1_G$ and $V \ni g_0$ (using G Hausdorff). Pick a continuous bump ϕ on U (Theorem 47), and consider $f := \phi$ as an element of $L^2(G)$ via $\iota_{C \rightarrow L^2}$. Then $\rho^R(g_0)f$ is the class of $h \mapsto \phi(hg_0)$, which has support in Ug_0^{-1} . Pick V small enough that $Vg_0^{-1} \cap U = \emptyset$ (possible by continuity of multiplication and Hausdorffness). Then $\rho^R(g_0)f$ vanishes on a neighbourhood of 1_G where f does not, so $\rho^R(g_0)f \neq f$ in $L^2(G)$. $\square$

Theorem 71 (Existence of separating self-adjoint convolution operator). *For every $g_0 \in G$ with $g_0 \neq 1_G$, there exists a symmetric bump $\phi \in C(G)$ such that:*

1. ϕ is symmetric (Definition 42);
2. T_ϕ is non-zero;
3. T_ϕ does not commute with $\rho^R(g_0)$ in the sense: there exists $f \in L^2(G)$ with $T_\phi(\rho^R(g_0)f - f) \neq 0$.

Proof. Pick $f \in L^2(G)$ with $f' := \rho^R(g_0)f - f \neq 0$ from Theorem 70. Pick a sequence of symmetric bumps ϕ_n on shrinking neighbourhoods of 1_G (Theorem 47, with the symmetry coming from the proof's symmetrization step). By Theorem 48, $T_{\phi_n}f' \rightarrow f' \neq 0$ in L^2 , so for some n , $T_{\phi_n}f' \neq 0$. Take $\phi := \phi_n$ for that n . $\square$

8.2 From convolution to a finite-dim invariant subspace

Theorem 72 (T_ϕ has a non-zero finite-dim eigenspace separating g_0). *For every $g_0 \neq 1_G$ in G , there exist $\phi \in C(G)$ symmetric, $\lambda \in \mathbb{R} \setminus \{0\}$, and a non-zero finite-dimensional closed subspace $W \leq L^2(G)$ such that:*

1. $W = \ker(T_\phi - \lambda \cdot \text{Id}) \neq 0$;
2. $\dim W < \infty$;
3. $\rho^R \text{.IsInvariant } W$;
4. $\rho^R(g_0)|_W \neq \text{Id}_W$.

Proof. Pick ϕ from Theorem 71. Then T_ϕ is compact (Theorem 45), self-adjoint (Corollary 44, using ϕ symmetric), and non-zero (item 2 of Theorem 71). By Theorem 50 there exists $\lambda \neq 0$ real and a finite-dim non-zero $H_\lambda := \ker(T_\phi - \lambda \text{Id})$. By Corollary 41 T_ϕ is a ρ^R -intertwiner; by Theorem 53 (eigenspace of an intertwiner is invariant), H_λ is ρ^R -invariant.

Now check (4). The full spectral decomposition (Assumption 49) gives $L^2(G) = \ker T_\phi \oplus \bigoplus_i H_{\lambda_i}$. By item 3 of Theorem 71, $T_\phi(\rho^R(g_0)f - f) \neq 0$, so the projection of $\rho^R(g_0)f - f$ onto $(\ker T_\phi)^\perp$ is non-zero, hence some component of $\rho^R(g_0)f - f$ in some H_{λ_i} is non-zero. Replacing λ by such λ_i and W by H_{λ_i} , we have $\rho^R(g_0)$ acting non-trivially on W . $\square$

8.3 From finite-dim invariant to finite-dim irreducible

Theorem 73 (Every finite-dim invariant subspace contains an irreducible). *Let W be a finite-dim non-zero closed subspace of H with $\rho \text{.IsInvariant } W$. Then there exists a non-zero closed subspace $W' \subseteq W$ with $\rho \text{.IsInvariant } W'$ such that $\rho|_{W'}$ is irreducible.*

Proof. Induction on $\dim W$. Base case $\dim W = 1$: W is itself irreducible. Inductive step: if $\rho|_W$ has a proper non-zero invariant subspace $W_1 \subsetneq W$, by Theorem 19 $W_1^\perp \cap W$ is also ρ -invariant of strictly smaller dimension. By induction one of the two pieces W_1 or $W_1^\perp \cap W$ contains an irreducible subspace; that is our W' . Otherwise $\rho|_W$ has only $\{0\}$ and W as invariant subspaces, so $\rho|_W$ itself is irreducible; take $W' := W$. $\square$

Theorem 74 (Finite-dim invariant non-trivial action splits to non-trivial irreducible action). *Let W be a finite-dim non-zero closed ρ -invariant subspace, and $g_0 \in G$ such that $\rho(g_0)|_W \neq \text{Id}_W$. Then there exists a non-zero closed ρ -invariant subspace $W' \subseteq W$ with $\rho|_{W'}$ irreducible and $\rho(g_0)|_{W'} \neq \text{Id}_{W'}$.*

Proof. By induction on $\dim W$, with the same splitting as in Theorem 73, but at each split step pick the side on which $\rho(g_0)$ still acts non-trivially. (At least one of the two pieces in $W = W_1 \oplus (W_1^\perp \cap W)$ has $\rho(g_0)$ non-trivial: if $\rho(g_0)$ acted as Id on both, it would act as Id on all of W , contradicting hypothesis.) Continue until reaching an irreducible subspace; since the property of “ $\rho(g_0)$ acts non-trivially” is preserved at each step, the resulting irreducible W' has $\rho(g_0)|_{W'} \neq \text{Id}$. $\square$

8.4 The separation theorem

Theorem 75 (Existence of separating finite-dim irreducible). *For every $g_0 \in G$ with $g_0 \neq 1_G$, there exists a non-zero finite-dim closed ρ^R -invariant subspace $W' \leq L^2(G)$ such that $\rho^R|_{W'}$ is irreducible and $\rho^R(g_0)|_{W'} \neq \text{Id}_{W'}$.*

Proof. Apply Theorem 72 to obtain a finite-dim invariant W with $\rho^R(g_0)|_W \neq \text{Id}_W$, then apply Theorem 74. $\square$

Chapter 9

Density of Matrix Coefficients

9.1 $\mathcal{M}$ separates points

Theorem 76 (Matrix coefficients of $\rho^R|_{W'}$ are matrix coefficients of G). *If $W' \leq L^2(G)$ is a finite-dim ρ^R -invariant subspace and $u, v \in W'$, then $\pi_{u,v}^{\rho^R|_{W'}}$ is a matrix coefficient of G in the sense of Definition 57, with the rep being $\rho^R|_{W'} : \text{UnitaryRep } G \ W'$.*

Proof. Tautological from the definitions: a matrix coefficient is a function on G , the rep is the restriction. $\square$

Theorem 77 ($\mathcal{M}$ separates points). *For every pair $g_1 \neq g_2$ in G there exists $f \in \mathcal{M}$ with $f(g_1) \neq f(g_2)$.*

Proof. Apply Theorem 75 with $g_0 := g_1 g_2^{-1}$ to get a finite-dim irreducible W' with $\rho^R(g_0)|_{W'} \neq \text{Id}$. Pick $u \in W'$ with $\rho^R(g_0)u \neq u$; pick $v \in W'$ with $\langle \rho^R(g_0)u - u, v \rangle \neq 0$. Then $\pi_{u,v}^{\rho^R|_{W'}}(g_0) \neq \pi_{u,v}^{\rho^R|_{W'}}(1_G)$. Right-translating: define $u' := \rho^R(g_2)u$. Then by the matrix-coefficient-translation identity $\pi_{u',v}^{\rho^R|_{W'}}(g) = \pi_{u,v}^{\rho^R|_{W'}}(gg_2)$, giving $\pi_{u',v}(g_1) \neq \pi_{u',v}(g_2)$. This is an element of $\mathcal{M}$ via Theorem 76. $\square$

Corollary 78 ($\mathcal{A}$ separates points). *$\mathcal{A}$ separates points (in the sense of `StarSubalgebra.SeparatesPoints`).*

Proof. $\mathcal{A} \supseteq \mathcal{M}$, and $\mathcal{M}$ separates points by Theorem 77; the larger set inherits the property. $\square$

9.2 Stone–Weierstrass: $\mathcal{A}$ is dense

Theorem 79 ($\mathcal{A}$ is uniformly dense in $C(G, \mathbb{C})$). *The closure of $\mathcal{A}$ in the uniform topology of $C(G, \mathbb{C})$ is the whole of $C(G, \mathbb{C})$.*

Proof. Apply `ContinuousMap.starSubalgebra_topologicalClosure_eq_top_of_separatesPoints` in `Mathlib.Topology.ContinuousMap.StoneWeierstrass`. Hypotheses: `[CompactSpace G]` (Definition 1); $\mathcal{A}$ a star-subalgebra (Definition 69); separates points (Corollary 78). $\square$

Corollary 80 ($\mathcal{A}$ is L^2 -dense in $L^2(G)$). *The image of $\mathcal{A}$ in $L^2(G)$ under $\iota_{C \rightarrow L^2}$ is dense in $L^2(G)$.*

Proof. $\mathcal{A}$ uniformly dense in $C(G)$ (Theorem 79) gives $\iota_{C \rightarrow L^2}(\mathcal{A})$ dense in $\iota_{C \rightarrow L^2}(C(G))$ in the L^2 topology (Theorem 13). And $\iota_{C \rightarrow L^2}(C(G))$ is dense in $L^2(G)$ (Theorem 12). $\square$

Chapter 10

Schur Orthogonality

10.1 Orthogonality across distinct irreducibles

Theorem 81 (Inner product of matrix coefficients across distinct irreducibles). *Let $\rho_1 : \mathbf{UnitaryRep} \ G \ V_1$, $\rho_2 : \mathbf{UnitaryRep} \ G \ V_2$ both finite-dimensional and irreducible with $\rho_1 \not\cong \rho_2$. Then for all $u_1, w_1 \in V_1$ and $u_2, w_2 \in V_2$,*

$$\int_G \pi_{u_1, w_1}^{\rho_1}(g) \overline{\pi_{u_2, w_2}^{\rho_2}(g)} d\mu_G(g) = 0.$$

Proof. Define $T : V_1 \rightarrow V_2$ (note: finite-dimensional) by $Tu := \int_G \pi_{u, w_1}^{\rho_1}(g) \cdot \rho_2(g^{-1})w_2 d\mu_G(g)$. Then T is bounded (finite-dimensional, hence automatic) and an intertwiner $\rho_1 \rightarrow \rho_2$ (verify by changing variables in the integral, using right-invariance). Compactness is automatic in finite dimensions. By Theorem 56, $T = 0$. Then $\langle Tu_1, u_2 \rangle_{V_2} = 0$, which expands to the stated integral being zero. $\square$

10.2 Same-irreducible orthogonality

Theorem 82 (Inner product of matrix coefficients of same irreducible). *Let $\rho : \mathbf{UnitaryRep} \ G \ V$ be finite-dimensional and irreducible, $d := \dim V$. Then for all $u_1, u_2, v_1, v_2 \in V$,*

$$\int_G \pi_{u_1, v_1}^{\rho}(g) \overline{\pi_{u_2, v_2}^{\rho}(g)} d\mu_G(g) = \frac{1}{d} \langle u_1, u_2 \rangle \overline{\langle v_1, v_2 \rangle}.$$

Proof. Define $T : V \rightarrow V$ by $Tu := \int_G \pi_{u, v_1}^{\rho}(g) \rho(g^{-1})v_2 d\mu_G(g)$ (finite-dimensional Bochner integral). T is an intertwiner $\rho \rightarrow \rho$, hence by Theorem 55 $T = c \cdot \text{Id}_V$ for some $c \in \mathbb{C}$. Take traces: $\text{tr}(T) = c \cdot d$, while $\text{tr}(T) = \int_G \pi_{v_1, v_1}^{\rho}(g) \cdot \text{tr}(\rho(g^{-1}) \cdot \langle \cdot, v_2 \rangle v_2) d\mu_G(g)$ evaluating the trace by an orthonormal basis $\{e_i\}$ of V gives $c \cdot d = \overline{\langle v_1, v_2 \rangle} \langle e_i, v_2 \rangle \dots$; carrying through the calculation yields $c = \overline{\langle v_1, v_2 \rangle} \langle u_1, u_2 \rangle / d$. $\square$

Chapter 11

Isotypic Decomposition of $L^2(G)$ and Plancherel

11.1 Isotypic component

Definition 83 (Set of finite-dim irreducible classes). $\widehat{G}$ is the type of equivalence classes (under $\cong$, Definition 26) of finite-dimensional continuous unitary irreducible representations of G . In Lean, $\widehat{G}$ is implemented as a quotient of the type of all finite-dim irreducibles modulo unitary equivalence, with a chosen representative V_ξ and rep ρ_ξ for each $[\xi]$.

Definition 84 (ξ -isotypic component of $L^2(G)$). For $\xi \in \widehat{G}$, the ξ -isotypic component is

$$L^2(G)_\xi := \overline{\sum \{V \mid V \text{ closed } \rho^R\text{-invariant subspace, } \rho^R|_V \cong \rho_\xi\}},$$

the closure (in L^2) of the linear span of all closed ρ^R -invariant subspaces of $L^2(G)$ on which ρ^R acts equivalently to ρ_ξ .

Theorem 85 ($L^2(G)_\xi$ is ρ^R -invariant). $\rho^R.IsInvariant L^2(G)_\xi$.

Proof. Sums of ρ^R -invariant subspaces are ρ^R -invariant; closures of ρ^R -invariant sets are ρ^R -invariant (since each $\rho^R(g)$ is continuous). $\square$

11.2 Orthogonality between isotypic components

Theorem 86 (Distinct isotypic components are orthogonal). For $\xi_1 \neq \xi_2$ in $\widehat{G}$, $L^2(G)_{\xi_1} \perp L^2(G)_{\xi_2}$.

Proof. By bilinearity and continuity of the inner product, it suffices to show $V_1 \perp V_2$ for any pair of closed ρ^R -invariant subspaces $V_1 \cong \rho_{\xi_1}$, $V_2 \cong \rho_{\xi_2}$. Consider the orthogonal projection $P : L^2(G) \rightarrow V_2$. It is bounded, ρ^R -equivariant (Theorem 19 plus the standard fact that orthogonal projection onto an invariant subspace is equivariant), and finite-rank (range is V_2 , finite-dim), hence compact. Restricted to V_1 , it gives a compact intertwiner $V_1 \rightarrow V_2$ between two non-isomorphic irreducibles. By Theorem 56, $P|_{V_1} = 0$, i.e. $V_1 \perp V_2$. $\square$

11.3 The isotypic Hilbert sum

Theorem 87 (Matrix coefficients of ρ_ξ live in $L^2(G)_\xi$). *For each $\xi \in \widehat{G}$ and $u, v \in V_\xi$, the matrix coefficient $\pi_{u,v}^{\rho_\xi} \in L^2(G)$ lies in $L^2(G)_\xi$.*

Proof. The closure of the right-translates of $\pi_{u,v}^{\rho_\xi}$ (under ρ^R) gives a closed ρ^R -invariant subspace $W \leq L^2(G)$. The map $V_\xi \rightarrow L^2(G)$, $u' \mapsto \pi_{u',v}^{\rho_\xi}$, is a G -equivariant linear map (right-translation action on $L^2(G)$ matches the ρ_ξ -action on V_ξ via the matrix-coefficient identity); it has trivial kernel (Schur, since V_ξ irreducible and the map non-zero), and image is W . Hence $\rho^R|_W \cong \rho_\xi$, so $W \subseteq L^2(G)_\xi$, hence $\pi_{u,v}^{\rho_\xi} \in W \subseteq L^2(G)_\xi$. $\square$

Theorem 88 (Closed sum of isotypic components is dense). $\overline{\sum_{\xi \in \widehat{G}} L^2(G)_\xi} = L^2(G)$.

Proof. By Theorem 87, every matrix coefficient lies in some $L^2(G)_\xi$. Hence $\mathcal{A} \subseteq \sum_{\xi} L^2(G)_\xi$. By Corollary 80, $\mathcal{A}$ is dense in $L^2(G)$, so the closed sum is all of $L^2(G)$. $\square$

Theorem 89 ($L^2(G)$ is the Hilbert sum of isotypic components). *The family $\{L^2(G)_\xi\}_{\xi \in \widehat{G}}$ is an `IsHilbertSum` structure on $L^2(G)$:*

- *it is an orthogonal family (Theorem 86);*
- *the closure of the linear span of all components is $\top$ (Theorem 88).*

Proof. Both conditions in the definition of `IsHilbertSum` (`Mathlib.Analysis.InnerProductSpace.l2Space`) are exactly the cited theorems. $\square$

11.4 Plancherel

Theorem 90 (Plancherel: per-isotypic norm sum). *For every $f \in L^2(G)$, $\|f\|_{L^2}^2 = \sum_{\xi \in \widehat{G}} \|P_\xi f\|_{L^2}^2$, where $P_\xi : L^2(G) \rightarrow L^2(G)_\xi$ is the orthogonal projection.*

Proof. Direct from `IsHilbertSum.norm_sq_eq_tsum` (or analogous Mathlib lemma) applied to Theorem 89. $\square$

Chapter 12

Mathlib-Status Summary

Counts (final v2 blueprint). 77 formal items across 9 chapters. Of these:

- **Mathlib citations ([ML]).** Haar measure (existence, uniqueness, invariance, inversion, finiteness on compact groups, probability normalization), `MeasureTheory.Lp`, L^2 inner product structure, `ContinuousMap.toLp` and density, L^p -monotonicity on finite measure spaces, `IsCompactOperator`, the closure of compact operators, complex Stone–Weierstrass, `IsHilbertSum`, `IsHilbertSum.norm_sq_eq_tsum`.
- **External assumption ([EXT]).** Exactly one item: the spectral theorem for compact self-adjoint operators on a complex Hilbert space (Assumption 49), plus its immediate consequence that a non-zero such operator has a non-zero finite-dimensional eigenspace (Theorem 50).
- **Trivial wrappers ([-ML]).** `PeterWeyl.haarProb` (rescaling), `cor:c2-MCA1g-L2-dense`, `thm:c2-plancherel-isotypic`.
- **New contributions ([NEW]).** Roughly 60 items, organized into:
 - *UnitaryRep API* (Chapter 2): structure definition, basic theorems about unitarity, subrep predicate and orthogonal-complement closure, irreducibility, intertwiners, equivalence, left/right regular representations.
 - *Convolution as a bounded operator on $L^2(G)$* (Chapter 3): pointwise convolution facts, bounded extension to L^2 , equivariance, self-adjointness for symmetric kernels, compactness.
 - *Approximate identities* (Chapter 4).
 - *Schur’s lemma* (Chapter 6, using Assumption 49).
 - *Matrix coefficients and the star-subalgebra* (Chapter 7).
 - *Existence of separating finite-dim irreducibles* (Chapter 8).
 - *Density of $\mathcal{A}$* (Chapter 9, the Stone–Weierstrass step).
 - *Schur orthogonality* (Chapter 10).
 - *Isotypic decomposition* (Chapter 11).

Translation-readiness checklist. Each item in the blueprint satisfies the following:

- Has a single named conclusion (no “ X is closed and invariant”; instead split into two atomic theorems).
- Has a `\lean{...}` directive giving the intended Lean name in a coherent namespace (`UnitaryRep.`, `PeterWeyl.`).
- Has typed signatures: every introduced object lists its type explicitly.
- Has a precise `\uses{...}` graph: each `\uses` entry is genuinely needed in the proof.
- Proofs decompose into named steps each of which corresponds to a single line/`have` in Lean. Where multiple Lean steps would be needed, those steps are extracted as their own theorems with their own `\uses` entries.